

5th Lab

Transfer Learning & Visualizations

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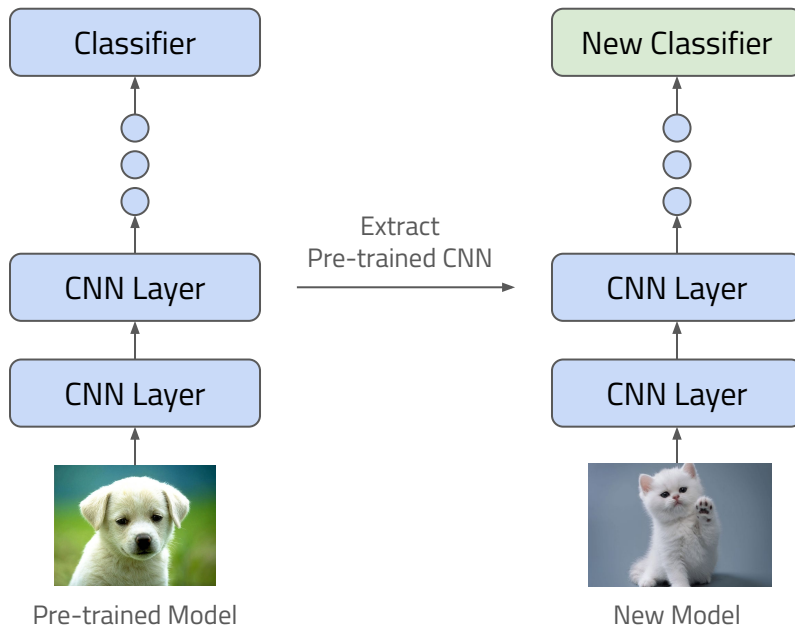


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Transfer Learning

- Most of the time, instead of creating and training a model from scratch, we can use **state-of-the-art models** (SoTA) that have been **pre-trained** on **large datasets**, such as ImageNet
- This process is called **transfer learning**

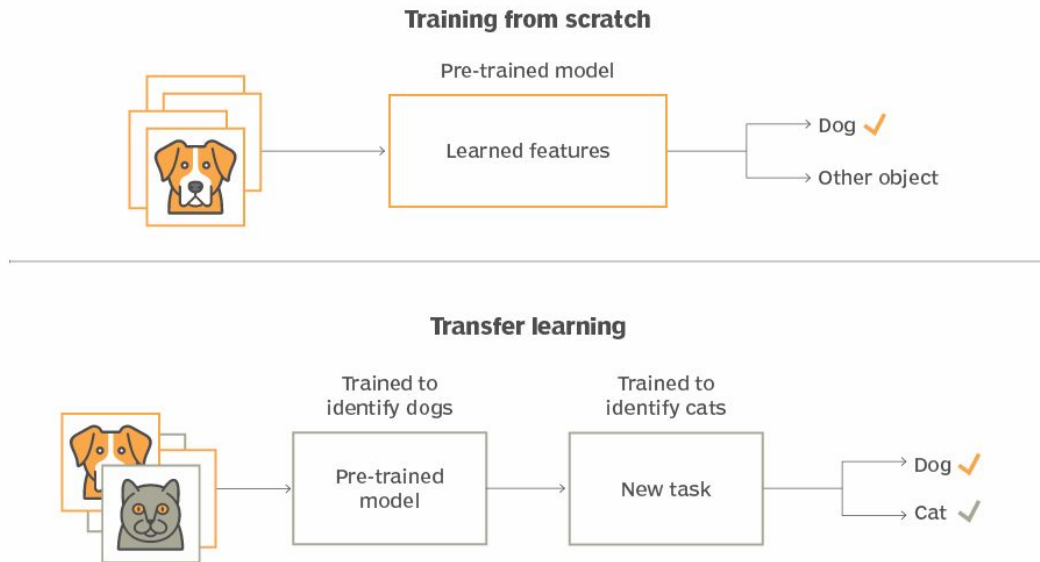


Transfer Learning

TL takes already-trained models or networks as a starting point. It then applies that model's knowledge gained in an initial source task or data towards a new, yet related, target task or data.

Advantages:

- Computational Cost
- Dataset Size
- Generalization



Transfer Learning

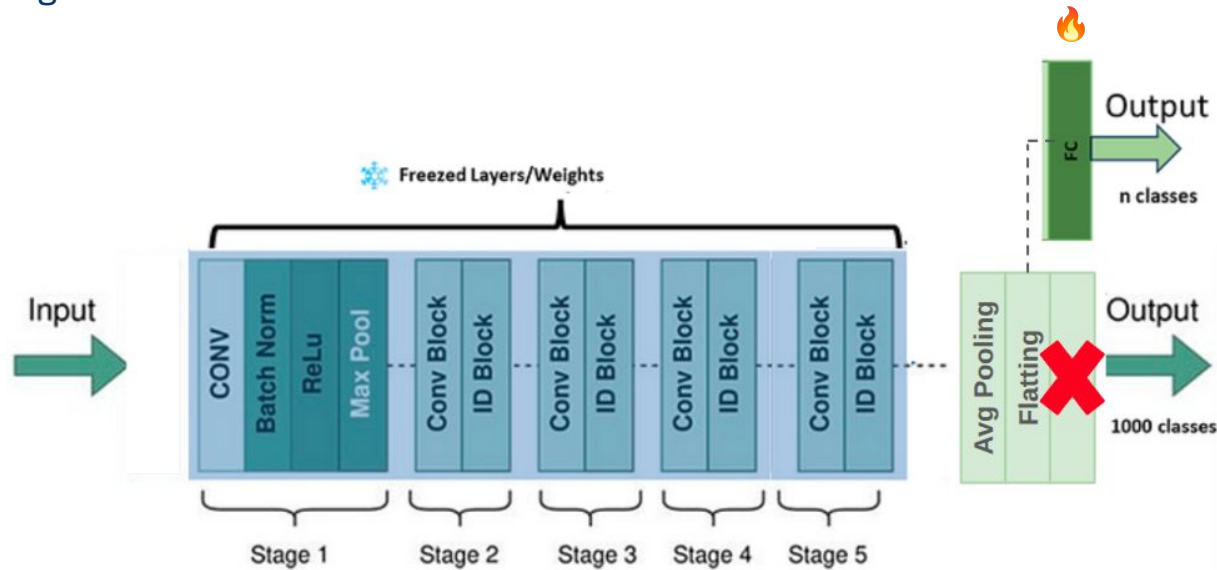
- Suppose we have a new dataset with 2 classes
- We want to **fine-tune on our dataset** the **ResNet** SoTA model, which has been trained on ImageNet

```
import torch.nn as nn
from torchvision.models import resnet18, ResNet18_Weights
...
N CLASSES = 2
...
model = resnet18(weights=ResNet18_Weights.IMAGENET1K_V1)
in_features = model.fc.in_features
model.fc = nn.Linear(in_features=in_features, out_features=N CLASSES)
```

- 1) Import ResNet model and weights
- 2) Define number of classes to use
- 3) Instantiate pre-trained ResNet model
- 4) Get the number of input features to classification layer
- 5) Instantiate a new classification layer

What are we really doing?

- ResNet is utilized as the foundational model to take advantage on its extensive pre-training.
- Preserving the weights of all original layers, except for those which are specifically adapted for training on a new dataset.



Hands-on with the Code!

- Open the notebook on Google Colab
- Download the dataset:
https://github.com/mrdbourke/pytorch-deep-learning/blob/main/data/pizza_steak_sushi.zip
- Solve the notebook
- Visualize the training loss and accuracy using Wandb or TensorBoard

Notebook @

<https://colab.research.google.com/drive/12F1MewpyfmssKbNEBIKDIYsDVRb2tYQ0?usp=sharing>

